



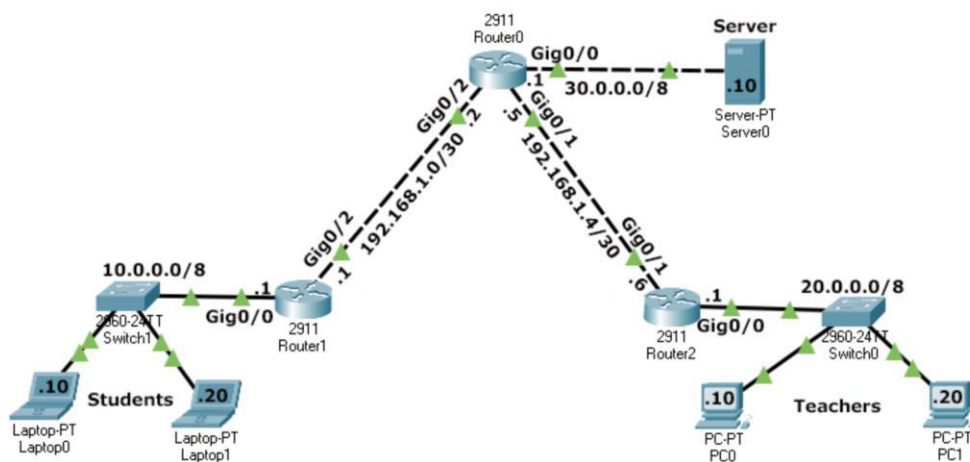
Computer Networks Lab
Fall 2025
Week 14

ACL and VLANS

Lab Task

Task 1 ACL:

1. Draw the **Network Topology** given below.
2. **Configure IP addresses** and routers interfaces according to the topology labelled below.
3. Configure **Static Routing** to the given network and test the connectivity by using **ping** command.
4. Create and implement a standard access list that blocks the **Students** section from accessing the **Server** section and **Teachers** section.
5. Create and implement a standard access list that permits only one of the Teachers PC to access the **Server** section and denies rest of the **Teachers** section.
6. Make sure to take all the necessary screenshots and submit them in a document along with the packet tracer file.



Task 2 VLANS:

The university is upgrading its internal network to improve organization and communication among different **Academic Units**. Each unit consists of a **Unit Supervisor (US)** and **four staff members**. The network team must separate these units using VLANs to ensure efficient connectivity within each unit while keeping units isolated from each other.

You are assigned to configure VLANs for the following **Academic Units**:

- **Electrical Unit (EU)**
- **Computing Unit (CU)**
- **Mechanical Unit (MU)**
- **Software Unit (SU)**

Each unit contains:

- **1 Unit Supervisor (US) computer**
- **4 Staff computers**

Your goal is to implement VLANs so that each Unit Supervisor can ping staff within their own unit, and each staff member can ping their own Unit Supervisor.